General Midterm Notes.

1. Interpolation
   (a) I recommend being comfortable with the Vandermonde matrix form.
   (b) I recommend being comfortable with cubic interpolation by hand (Lagrange, Hermite, Spline)
2. Lagrange Interpolating Polynomials
   (a) I recommend learning and using the $\Psi(x)$ representation.
3. Divided Differences
   (a) You should be familiar with the divided difference representation of the interpolation schemes.
   (b) You should be familiar with the notion of derivatives and taking derivatives in the divided difference notation.
4. Splines
   (a) You should be familiar with cubic splines and their advantages.
5. Numerical Differentiation
   (a) You should be familiar with how to compute a three point scheme.
   (b) You should be familiar with the advantages of centered schemes versus end point schemes.
   (c) You should be familiar with using Ψ notation and the implications when $\Psi(x) = 0$ and when $\Psi'(x) = 0$.
   (d) You should have some intuition about the consequence of round off errors as $h \to 0$.
6. Numerical Integration
   (a) You should be familiar with the process for computing coefficients for numerical integration.
   (b) You should be familiar with when the integral mean value theorem applies and when it does not (and what to do in this case).

Rote Level Questions

1. You will be expected to know how to formulate Lagrange interpolating polynomials, Hermite interpolating polynomials and cubic splines (and the general definition of a spline).
2. You will be provided exact error expressions.
3. You will be provided exact expressions for Ψ' or integrals of Ψ.

Conceptual Level Questions

1. Emphasis on connections to linear algebra and the degrees of freedom in a system. How can you represent interpolating schemes as a system of linear equations?
2. Emphasis on connections to calculus and the second derivative for oscillatory behaviors. Can you construct examples where high degree interpolation schemes oscillate wildly, and show that cubic splines improve performance?
3. Emphasis on connections to round off errors and their impact on taking smaller step sizes. Why would I expect the error to decrease as I decrease the mesh size, and then why can the error even increase as I decrease the mesh size?

Vandermonde formulation. Suppose we seek a polynomial

$$p(x) = a_0 + a_1 x + \cdots + a_m x^m.$$

Since (3.1.1) imposes $n+1$ conditions on $p(x)$, it is reasonable to first consider the case $m = n$. Then we want to find $a_0, a_1, \ldots, a_n$ such that

$$\begin{aligned} a_0 + a_1 x_0 + a_2 x_0^2 + \cdots + a_n x_0^n &= y_0, \\ &\vdots \\ a_0 + a_1 x_n + a_2 x_n^2 + \cdots + a_n x_n^n &= y_n. \end{aligned}$$

This is a system of $n+1$ linear equations in $n+1$ unknowns, completely equivalent to solving the polynomial interpolation problem. In vector–matrix notation, the system is

$$X\, a = y$$

with

$$X_{i,j} = [x_i^j], \qquad i, j = 0, 1, \ldots, n,$$

$$a = \begin{bmatrix} a_0 & a_1 & \cdots & a_n \end{bmatrix}^\top, \qquad y = \begin{bmatrix} f(x_0) & \cdots & f(x_n) \end{bmatrix}^\top.$$

The matrix X is called a *Vandermonde matrix*.

Example (Cubic Hermite interpolation). The most widely used form of Hermite interpolation is the cubic Hermite polynomial, which solves

$$p(a) = f(a), \qquad p'(a) = f'(a), \qquad p(b) = f(b), \qquad p'(b) = f'(b). \tag{1}$$

The divided–difference form becomes

$$H_2(x) = f(a) + (x-a) f'(a) + (x-a)^2 f[a,a,b] + (x-a)^2 (x-b) f[a,a,b,b], \tag{2}$$

in which

$$f[a,a,b] = \frac{f[a,b] - f'(a)}{b-a}, \qquad f[a,a,b,b] = \frac{f'(b) - 2f[a,b] + f'(a)}{(b-a)^2}.$$

The error formula is

$$\begin{aligned} f(x) - H_2(x) &= (x-a)^2 (x-b)^2\, f[a,a,b,b,x] \\ &= \frac{(x-a)^2 (x-b)^2}{24}\, f^{(4)}(\xi_x), \end{aligned} \tag{3}$$

and hence

$$\max_{a \le x \le b} |f(x) - H_2(x)| \;\le\; \frac{(b-a)^4}{384} \max_{a \le t \le b} |f^{(4)}(t)|. \tag{4}$$

Notation. For a piecewise polynomial function $p(x)$, there is an associated grid

$$-\infty < x_0 < x_1 < \cdots < x_n < \infty. \tag{5}$$

The points x_j are sometimes called *knots*, *breakpoints*, or *nodes*. The function $p(x)$ is a polynomial on each of the subintervals

$$(-\infty, x_0],\; [x_0, x_1],\; \ldots,\; [x_{n-1}, x_n],\; [x_n, \infty), \tag{6}$$

although often the end intervals $(-\infty, x_0]$ and $[x_n, \infty)$ are not included. We say $p(x)$ is a *piecewise polynomial of order* r if the degree of $p(x)$ is at most r on each of the subintervals. Unless specified otherwise, no continuity across the knots is required.

Spline functions. As before, consider a grid

$$a = x_0 < x_1 < \cdots < x_n = b.$$

We say $s(x)$ is a *spline function of order* $m \geq 1$ if it satisfies:

(P1) On each subinterval $[x_{i-1}, x_i]$, $s(x)$ is a polynomial of degree $< m$.

(P2) For $0 \leq r \leq m-2$, the derivative $s^{(r)}(x)$ is continuous on $[a, b]$ (i.e., $s \in C^{m-2}[a, b]$).

The derivative of a spline of order m is a spline of order $m-1$ (and similarly antidifferentiation raises the order by one). Moreover, if the continuity in (P2) is strengthened to include $s^{(m-1)}(x)$ (i.e., $s \in C^{m-1}[a, b]$), then $s(x)$ must in fact be a single polynomial on $[a, b]$ of degree $\leq m-1$.

Degrees of Freedom. For our interpolation problem, we seek a *cubic spline* $s(x)$ such that

$$s(x_i) = y_i, \qquad i = 0, 1, \dots, n. \tag{7}$$

On each subinterval $[x_{i-1}, x_i]$ we write

$$s(x) = a_i + b_i x + c_i x^2 + d_i x^3, \qquad i = 1, \dots, n. \tag{8}$$

There are $4n$ unknown coefficients $\{a_i, b_i, c_i, d_i\}$. The constraints are the interpolation conditions $(3.7.11)$ together with the spline continuity conditions (from C^2-continuity):

$$s^{(j)}(x_i + 0) = s^{(j)}(x_i - 0), \qquad i = 1, \dots, n-1, \quad j = 0, 1, 2. \tag{9}$$

Counting equations gives

$$(n+1) + 3(n-1) = 4n - 2,$$

compared with $4n$ unknowns. Hence two additional conditions (boundary conditions) are required to determine a unique cubic interpolating spline $s(x)$.

Oscillatory Behaviors. Let $f \in C^4[a, b]$ and let $x_0 = a < x_1 < \cdots < x_n = b$ be a partition. Let s_c be the *complete (clamped) cubic spline* interpolant of f satisfying

$$s_c(x_i) = f(x_i) \quad (i = 0, \dots, n), \qquad s_c'(a) = f'(a), \;\; s_c'(b) = f'(b).$$

Consider the admissible class

$$\mathcal{A} := \{\, g \in C^2[a, b] : \; g(x_i) = f(x_i) \;\; (i = 0, \dots, n), \; g'(a) = f'(a), \; g'(b) = f'(b) \,\}.$$

Then

$$\int_a^b |s_c''(x)|^2 \, dx \;\leq\; \int_a^b |g''(x)|^2 \, dx, \qquad \forall\, g \in \mathcal{A}, \tag{10}$$

with equality iff $g \equiv s_c$.

Proof. Let $g \in \mathcal{A}$ and set $k := s_c - g$. Then

$$k(x_i) = 0 \ (i = 0, \dots, n), \qquad k'(a) = k'(b) = 0.$$

Expanding the square,

$$\int_a^b |g''|^2 = \int_a^b |s_c'' - k''|^2 = \int_a^b |s_c''|^2 - 2\int_a^b s_c'' k'' + \int_a^b |k''|^2.$$

Using integration by parts one can find

$$\int_a^b s_c''(x)\, k''(x)\, dx = 0. \tag{11}$$

Indeed, integrate by parts on each subinterval $[x_{i-1}, x_i]$ and sum:

$$\int_{x_{i-1}}^{x_i} s_c'' k''\, dx = \Big[s_c'' k' - s_c''' k \Big]_{x_{i-1}}^{x_i} - \int_{x_{i-1}}^{x_i} s_c^{(4)} k\, dx.$$

Since s_c is cubic on each $[x_{i-1}, x_i]$, $s_c^{(4)} \equiv 0$ there. Summing over $i = 1, \dots, n$ gives telescoping boundary terms. Using that s_c'' is continuous at each knot, $k(x_i) = 0$ for all i, and $k'(a) = k'(b) = 0$, all boundary contributions vanish; hence (**??**).

Therefore,

$$\int_a^b |g''|^2 = \int_a^b |s_c''|^2 + \int_a^b |k''|^2 \ \geq \ \int_a^b |s_c''|^2. \tag{12}$$

Equality holds iff $k'' \equiv 0$, i.e. k is linear. The interpolation and clamped boundary conditions then force $k \equiv 0$, so $g \equiv s_c$. □

Exploring *not-a-knot* condition.

Theorem 1 (Error of the complete (clamped) cubic spline). *Let $f \in C^4[a,b]$. For each n, let a partition*

$$\tau_n : \quad a = x_0^{(n)} < x_1^{(n)} < \cdots < x_n^{(n)} = b,$$

and define the mesh width

$$|\tau_n| = \max_{1 \leq i \leq n} (x_i^{(n)} - x_{i-1}^{(n)}).$$

Let $s_{c,n}$ be the complete (clamped) cubic spline *interpolant of f on τ_n, i.e.*

$$s_{c,n}\left(x_i^{(n)}\right) = f\left(x_i^{(n)}\right) \quad (i = 0, 1, \dots, n), \qquad s_{c,n}'(a) = f'(a), \ \ s_{c,n}'(b) = f'(b).$$

Then there exist absolute constants c_j such that for $j = 0, 1, 2$,

$$\max_{a \leq x \leq b} \left| f^{(j)}(x) - s_{c,n}^{(j)}(x) \right| \ \leq \ c_j \, |\tau_n|^{4-j} \max_{a \leq x \leq b} |f^{(4)}(x)|. \tag{13}$$

Acceptable constants are

$$c_0 = \frac{5}{384}, \qquad c_1 = \frac{1}{24}, \qquad c_2 = \frac{3}{8}.$$

Numerical Differentiation

One of the main approaches to deriving a numerical approximation to $f'(x)$ is to use the derivative of a polynomial $p_n(x)$ that interpolates $f(x)$ at a given set of node points. Let $x_0, x_1, \ldots, x_n$ be given, and let $p_n(x)$ interpolate $f(x)$ at these nodes. Usually $\{x_i\}$ are evenly spaced.

Key Definitions

$$f'(x) \approx p_n'(x).$$

$$p_n(x) = \sum_{j=0}^{n} f(x_j)\, l_j(x),$$

$$l_j(x) = \frac{\Psi_n(x)}{(x - x_j)\, \Psi_n'(x_j)},$$

$$l_j(x) = \frac{(x-x_0)\cdots(x-x_{j-1})(x-x_{j+1})\cdots(x-x_n)}{(x_j-x_0)\cdots(x_j-x_{j-1})(x_j-x_{j+1})\cdots(x_j-x_n)},$$

$$\Psi_n(x) = (x-x_0)\cdots(x-x_n),$$

$$f(x) - p_n(x) = \Psi_n(x)\, f[x_0, \ldots, x_n, x].$$

Derivative Approximation using Interpolation Thus

$$f'(x) \approx p_n'(x) = \sum_{j=0}^{n} f(x_j)\, l_j'(x) \;\equiv\; D_h f(x).$$

$$f'(x) - p_n'(x) = \Psi_n'(x)\, f[x_0, \ldots, x_n, x] + \Psi_n(x)\, f[x_0, \ldots, x_n, x, x].$$

From the definitions above

$$f'(x) - p_n'(x) = \Psi_n'(x)\, \frac{f^{(n+1)}(\xi_1)}{(n+1)!} + \Psi_n(x)\, \frac{f^{(n+2)}(\xi_2)}{(n+2)!},$$

with $\xi_1, \xi_2 \in \mathcal{X}\{x_0, \ldots, x_n, x\}$. Higher order differentiation formulas and their error can be obtained by further differentiation.

The most common application of the preceding is to evenly spaced nodes $\{x_i\}$. Thus let

$$x_i = x_0 + ih, \qquad i \geq 0,$$

with $h > 0$. In this case, it is straightforward to show that

$$\Psi_n(x) = \mathcal{O}(h^{n+1}), \qquad \Psi_n'(x) = \mathcal{O}(h^n).$$

Thus

$$f'(x) - p_n'(x) = \begin{cases} \mathcal{O}(h^n) & \Psi_n'(x) \neq 0, \\ \mathcal{O}(h^{n+1}) & \Psi_n'(x) = 0. \end{cases}$$

Example 1. Let $n = 1$, so that $p_n(x)$ is just the linear interpolate of $(x_0, f(x_0))$ and $(x_1, f(x_1))$. Then (5.7.3) yields

$$f'(x_0) \doteq p_n'(x_0) \;\equiv\; \frac{1}{h}\,[\,f(x_0+h) - f(x_0)\,].$$

$$f'(x_0) - p_n'(x_0) = \frac{h}{2}\,f''(\xi_1), \qquad x_0 \le \xi_1 \le x_1,$$

since $\Psi(x_0) = 0$.

Example 2. To improve on this with linear interpolation, choose $x = m \equiv (x_0 + x_1)/2$. Then

$$f'(m) \doteq \frac{1}{h}\,[\,f(x_1) - f(x_0)\,].$$

We usually rewrite this by letting $\delta = h/2$, to obtain

$$f'(m) \doteq D_\delta f(m) = \frac{1}{2\delta}\,[\,f(m+\delta) - f(m-\delta)\,].$$

For the error,

$$f'(m) - D_\delta f(m) = -\,\frac{\delta^2}{6}\,f^{(3)}(\xi_2), \qquad m - \delta \le \xi_2 \le m + \delta.$$

In general, to obtain the higher-order case, we want to choose the nodes $\{x_i\}$ to have $\Psi_n'(x) = 0$. This will be true if n is odd and the nodes are placed symmetrically about x.
To obtain higher-order formulas in which the nodes all lie on one side of x, use higher values of n. For example, with $x = x_0$ and $n = 2$,

$$f'(x_0) \doteq D_h f(x_0) \equiv \frac{1}{2h}\,[-3f(x_0) + 4f(x_1) - f(x_2)],$$

$$f'(x_0) - D_h f(x_0) = \frac{h^2}{3}\,f^{(3)}(\xi_1), \qquad x_0 \le \xi_1 \le x_2.$$

Aitken Extrapolation for Sequences

Reconsider

$$\lim_{n\to\infty} \frac{x - x_{n+1}}{x - x_n} \;=\; g'(x).$$

for a convergent iteration

$$x_{n+1} = g(x_n), \qquad n \ge 0.$$

Considering only linear convergent sequences.
Thus we will assume

$$0 < |g'(x)| < 1.$$

We examine estimating the error in the iterates and give a way to accelerate the convergence of $\{x_n\}$. We begin by considering the ratios

$$\lambda_n = \frac{x_n - x_{n-1}}{x_{n-1} - x_{n-2}}, \qquad n \ge 2.$$

Claim:

$$\lim_{n\to\infty} \lambda_n = g'(x).$$

To see this, write

$$\lambda_n = \frac{(x - x_{n-1}) - (x - x_n)}{(x - x_{n-2}) - (x - x_{n-1})}.$$

Then

$$\lambda_n = \frac{(x - x_{n-1}) - g'(\xi_{n-1})(x - x_{n-1})}{(x - x_{n-2})/[\,g'(\xi_{n-2})\,] - (x - x_{n-1})} = \frac{1 - g'(\xi_{n-1})}{1/[\,g'(\xi_{n-2})\,] - 1}.$$

Hence

$$\lim_{n\to\infty} \lambda_n = \frac{1 - g'(x)}{1/[\,g'(x)\,] - 1} = g'(x).$$

The quantity λ_n is computable, and when it converges empirically to a value λ, we assume $\lambda = g'(x)$. We use $\lambda_n \approx g'(x)$ to estimate the error in the iterates x_n. Assume

$$x - x_n \doteq \lambda_n (x - x_{n-1}).$$

Then

$$x - x_n = (x - x_{n-1}) + (x_{n-1} - x_n) \doteq \frac{1}{\lambda_n}(x - x_n) + (x_{n-1} - x_n),$$

so that

$$x - x_n \doteq \frac{\lambda_n}{1 - \lambda_n}(x_n - x_{n-1})$$

The Trapezoidal Rule

The simple trapezoidal rule is based on approximating $f(x)$ by the straight line joining $(a, f(a))$ and $(b, f(b))$. By integrating the formula for this straight line, we obtain the approximation

$$I_1(f) = \left(\frac{b-a}{2}\right)[f(a) + f(b)]$$

To obtain an error formula, we use the interpolation error formula

$$f(x) - \frac{(b-x)f(a) + (x-a)f(b)}{b-a} = (x-a)(x-b)f[a, b, x]$$

We also assume for all work with the error for the trapezoidal rule in this section that $f(x)$ is twice continuously differentiable on $[a, b]$. Then

$$\begin{aligned} E_1(f) &= \int_a^b f(x)\,dx - \frac{(b-a)}{2}[f(a) + f(b)] \\ &= \int_a^b (x-a)(x-b)f[a, b, x]\,dx \end{aligned}$$

Using the integral mean value theorem.

$$\begin{aligned} E_1(f) &= f[a, b, \xi]\int_a^b (x-a)(x-b)\,dx, \quad \text{for some } a \le \xi \le b \\ &= \left[\frac{1}{2}f''(\eta)\right]\left[-\frac{1}{6}(b-a)^3\right], \quad \text{for some } \eta \in [a, b]. \end{aligned}$$

Simpson's rule

To improve upon the simple trapezoidal rule, we use a quadratic interpolating polynomial $p_2(f)$ to approximate $f(x)$ on $[a,b]$. Let $c=(a+b)/2$, and define

$$I_2(f)=\int_a^b\left[\frac{(x-c)(x-b)}{(a-c)(a-b)}\,f(a)+\frac{(x-a)(x-b)}{(c-a)(c-b)}\,f(c)+\frac{(x-a)(x-c)}{(b-a)(b-c)}\,f(b)\right]dx.$$

Carrying out the integration, we obtain

$$I_2(f)=\frac{h}{3}\left[f(a)+4\,f\left(\frac{a+b}{2}\right)+f(b)\right],\qquad h=\frac{b-a}{2}.$$

This is called *Simpson's rule*. An illustration is given in Figure 5.2, with the shaded region denoting the area under the graph of $y=p_2(x)$.
For the error, we begin with the interpolation error formula (3.2.11) to obtain

$$E_2(f)=I(f)-I_2(f)$$

We cannot apply the integral mean value theorem since the polynomial in the integrand changes sign at $x=c=(a+b)/2$. We will assume $f(x)$ is four times continuously differentiable on $[a,b]$ for the work of this section on Simpson's rule. Define

$$w(x)=\int_a^x(t-a)(t-c)(t-b)\,dt.$$

It is not hard to show that

$$w(a)=w(b)=0,\qquad w(x)>0\quad\text{for } a<x<b.$$

Integrating by parts,

$$\begin{aligned}E_2(f)&=\int_a^b w'(x)\,f[a,b,c,x]\,dx\\&=\big[\,w(x)\,f[a,b,c,x]\,\big]_{x=a}^{x=b}-\int_a^b w(x)\,\frac{d}{dx}f[a,b,c,x]\,dx\\&=-\int_a^b w(x)\,f[a,b,c,x,x]\,dx.\end{aligned}$$

Applying the integral mean value theorem

$$\begin{aligned}E_2(f)&=-\,f[a,b,c,\xi,\xi]\int_a^b w(x)\,dx,\qquad\text{for some } a\le\xi\le b\\&=-\frac{f^{(4)}(\eta)}{24}\left[\frac{4}{15}h^5\right],\qquad h=\frac{b-a}{2},\quad\text{for some } \eta\in[a,b].\end{aligned}$$

Thus

$$E_2(f)\;=\;-\frac{h^5}{90}\,f^{(4)}(\eta),\qquad \eta\in[a,b].$$